

SECTION 15810-A

HVAC SHEET METAL DUCTWORK

PART 1 - GENERAL

1.1 SUMMARY

- A. Sections includes sheet metal materials, fasteners, supports and duct construction classification for supply, return, and exhaust systems.

1.2 DEFINITIONS

- A. Ductwork sizes on Drawings state inside clear dimensions. For acoustically lined and internally insulated ductwork, maintain ductwork sizes inside lining or insulation.

1.3 REFERENCES, CODES, STANDARDS

- A. Construct ductwork in accordance with referenced codes and standards as listed below, unless herein modified and limited otherwise.
 - 1. 1985 SMACNA First Edition, HVAC Duct Construction Standards, Metal and Flexible
 - 2. 1992 SMACNA Round and Rectangular Industrial Duct Construction Standards
 - 3. 1993 SMACNA Accepted Industry Practice for Industrial Duct Construction
 - 4. 1991 SMACNA First Edition, Seismic Restraint Manual Guidelines for Mechanical Systems, with California Addendum E.
 - 5. 1997 Uniform Mechanical Code, Chapter 6, Duct Systems
 - 6. 1997 Uniform Mechanical Code Standard 6-2, Standard for Metal Ducts
 - 7. 1997 UMC Chapter 5, Exhaust Systems
 - 8. 1996 NFPA, 90-A, Standard for Installation of Air Conditioning and Ventilating Systems
 - 9. 1996 NFPA 96, Ventilation Control and Fire Protection of Commercial Cooking Operators
 - 10. 1996 NFPA 91, Installation of Exhaust Systems for Air Conveying of Materials
 - 11. 1993 ASHRAE, Chapter 32, Duct Design
 - 12. 1995 ASHRAE, Chapter 26, Industrial Exhaust Systems
 - 13. 1992 ASHRAE, Chapter 16, Duct Construction
 - 14. 1992 21st Edition ACGIH Industrial Ventilation Manual of Recommended Practice
 - 15. Sheet Metal Welding Code ANSI/AWS D9.1-90
 - 16. ASTM A90, Test Method for Weight of Coating on Zinc Coated Iron or Steel Articles

17. ASTM A167, Stainless and Heat-Resistant Chromium-Nickel Steel Plate, Sheet, and Strip
18. ASTM A366, Steel, Sheet, Carbon, Cold Rolled Commercial Quality
19. ASTM A480, Flat-Pulled Stainless and Heat-Resistant Steel Plate, Sheet and Strip
20. ASTM A568, Steel, Sheet, Carbon and High Strength Low Alloy; Hot Rolled and Cold Rolled
21. ASTM A569, Steel, Carbon, Hot Rolled Sheet and Strip Commercial Quality
22. ASTM A653, Steel Sheet, Zinc Coated (Galvanized) or Zinc Iron Alloy Coated (Galvanealed) by the Hot-Dip Process
23. ASTM A700, Practices for Packing, Marking and Loading Methods for Steel Products for Domestic Shipment
24. ASTM A924, General Requirements for Sheet Steel, Metallic Coated by the Hot-Dip Process

1.4 DUCT CONSTRUCTION CRITERIA MODIFICATION AND LIMITATIONS

- A. Formed-on flanges shall be constructed as a SMACNA T-24 flange, whose limits are defined on Page 1-25 1985 SMACNA Manual, First Edition. Formed-on flanges shall only be accepted for use on ductwork 42 inches wide or less and subjected to 2 inch w.g.s.p. or less. No other construction pertaining to formed-on flanges, will be permitted. Formed-on flanges TDC/TDF/T-24A/T-25A/T-25B shall not be permitted.
- B. Construct round and rectangular exhaust system ductwork 2 inch w.g. static negative or greater, in accordance with applicable classification of 1992 SMACNA industrial duct standards. Refer to Section 15843.
- C. Construct rectangular return systems ductwork 3 inch w.g. static negative or greater, in accordance with 1992 SMACNA Class 1 industrial duct standards. Refer to Section 15843.
- D. Construct round return systems ductwork 2 inch w.g. static negative or greater, in accordance with 1992 SMACNA Class 1 industrial duct standards. Refer to Section 15843.
- E. Oval ductwork shall not be used for negative duct systems.
- F. When SMACNA 1985 First Edition is applicable, rectangular duct intermediate reinforcement shall be in accordance with Tables 1-3 through 1-19.
- G. Transverse joints and intermediate reinforcements shall

maintain their structural integrity within the operating range of the system. No duct deformation or deflection which exceeds 1985 SMACNA Manual First Edition deflection criteria will be permitted.

- H. SMACNA 1985 First Edition tie rod option construction, Page 1-27, shall not be permitted on negative pressure duct applications.
- I. The following HVAC duct construction means and methods shall not be permitted.
 - 1. 1995 SMACNA Second Edition in its entirety
 - 2. Proprietary system equivalency procedures discussed under Section 7 of 1985 SMACNA First Edition
 - 3. Conduit type tie-rods with threaded friction inserts
 - 4. Snap lock longitudinal seams on 2 inch w.g. static positive or greater ductwork
 - 5. Tie-rod option reinforcement on negative systems
 - 6. Contractor's duct construction tables not in compliance with 1985 SMACNA Tables 1 through 13 or UMC Tables A6-2-A through A6-2-P.
 - 7. PVC clips on flange systems. PVC clips may be used only for listed fire damper break-away connections.
- J. Proprietary duct construction means shall be permitted only as follows:
 - 1. Slide-on connectors Ductmate 35, Ward J., or WDCI flange systems with metal clips and bolted corners. Components shall be manufactured by a single source. Each component shall be stamped with manufacturer's name or mark. PVC clips are not allowed.
 - 2. Duct shall be constructed using 1985 SMACNA First Edition, Construction Tables 1 through 13. Proprietary system's duct construction tables and methods, will not be permitted.
 - 3. Tie-rods to be minimum 3/8 inch galvanized rod. Rod to be attached to the duct wall or to a reinforcement and the duct wall, using internal and external nuts. The attachment of tie-rods by welding or bolting is also permitted.

1.5 SUBMITTALS

- A. Submit in accordance with Division 1.
- B. Submit typical shop standards and/or SMACNA details for each class of duct specified, including particulars such as gauge sizes, welds, joint details, and fitting configurations prior to start of work.
- C. Submit written report confirming ductwork has been

fabricated and installed in accordance with SMACNA Standards.

- D. Submit shop drawings of ductwork only when it is necessary to deviate from the intent of the design Drawings. Minor changes in routing or duct aspect ratio to avoid field conflicts do not require shop drawing submittals.

1.6 DIAGRAMS

- A. Construction Documents may include diagrams as a part of the working Drawings. These diagrams are not for the purpose of giving physical dimensions or locations but rather to identify system components, controls and to make clear the interconnections, by the duct systems, of the various units of the process. If an item is shown on either the diagram drawings or the plan drawings, but not on both, it will be assumed that the contractor has included such item in his estimate of the cost of the work and that he shall provide same.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Manufactured Round and Flat Oval Ductwork Including Preinsulated Double Wall and Acoustical Duct:
 - 1. According to the recommended manufacturer table.
- B. Acoustical Plenums:
 - 1. According to the recommended manufacturer table.
- C. Proprietary Manufactured Duct Joints Rectangular, Round, Flat Oval:
 - 1. According to the recommended manufacturer table.

2.2 MATERIALS

- A. Galvanized Ductwork: Fabricated using galvanized steel sheet of lock forming quality, having zinc coating of 1.25 ounces per square foot for each side per ASTM A653, G 90. All ductwork, supports, and reinforcement shall be galvanized unless otherwise noted.
 - 1. Carbon Steel Sheets: ASTM A 366, cold-rolled sheets, commercial quality, with oiled, exposed matte finish.
 - 2. Stainless Steel: ASTM A 480, Type 316, sheet form, with No. 4 finish on exposed surface for ducts exposed to view; Type 304, sheet form, with No. 1 finish for concealed ducts.

- B. Reinforcement Shapes and Plates: Unless otherwise indicated, provide galvanized steel reinforcing where installed on galvanized sheet metal ducts. For aluminum and stainless steel ducts provide reinforcing of compatible materials.
- C. PVC Coated Sheet Metal: Fabricated using galvanized steel sheet as specified above with a coating of polyvinyl chloride of not less than 1 mils thickness external and not less than 4 mils thickness internal for laboratory exhaust and not less than 4 mils thickness external and not less than 1 mil thickness internal for buried ductwork.
- D. Support Materials: Refer to Sections 15245 and 15140.
- E. Fasteners: Use galvanized blind rivets and bolts throughout; sheet metal screws are acceptable on low pressure ducts only.
- F. Proprietary Duct Joint Systems: Components consisting entirely of metal components. PVC components will not be acceptable.
- G. Sealant Material: Duct sealer shall be flexible, water-based, adhesive sealant designed for use in all pressure duct systems. After curing, it shall be resistant to ultraviolet light and shall seal out water, air, and moisture. Sealer shall be UL listed and conform to ASTM E 84.
- H. Joints, Seams:
 - 1. All joints shall be in accordance with Referenced Code and Standards based upon size and pressure classification except as modified.
 - 2. Welded joints shall be in accordance with the Welding Sheet Metal Code.
 - 3. Manufactured joints, where applicable may be used as herein specified.
 - 4. Make up all laboratory exhaust system ductwork joints with Racheem Thermofit wrap around duct bands installed in strict accordance with manufacturer's instructions.
- I. Flanged Interior Gasket:
 - 1. Ductmate 440 or a Butyl Rubber Gasket which meets MIL-C 18969B, Type II Class B, TT-C-1796 A, Type II Class B, and TTS-S-001657 must also pass UL 723.

2.3 DUCT CONSTRUCTION CLASSIFICATIONS

- A. General: Construct and seal ductwork in accordance with SMACNA pressure classifications and seal classes listed for ductwork systems involved.

System Description	Pressure Class, "W.G."	Seal Class	Duct Material	Reference Spec. Section
Supply Air Ducts (upstream terminal units)	+3	A	GSM	15840
(downstream terminal units or reheat coils)	+1	C	GSM	15840
Return Air Ducts	-2	B	GSM	15840
Common Exhaust Air Ducts	-1	B	GSM	15840
Outside Air Ducts	-1	B	GSM	15840
Kitchen	-2	W	SS	15840
KEY FOR THE ABOVE: W= ALL WELDED CONSTRUCTION. S= SHRINK FIT COUPLING GSM= GALVANIZED SHEET METAL. PVS= PVC COATED SHEET METAL.				

2.4 SHEET METAL PLENUMS

- A. Plenum: Two inch insulated double-wall plenums shall be fabricated and erected as shown in SMACNA 1985 Standards, Chapter 6. Where plenums extend down to finished floor, plenums shall be terminated at the curb line and anchored by the use of galvanized angle iron sealed and bolted to the curb.

2.5 ACOUSTICAL PLENUMS

- A. General: Pre-engineered, prefabricated, double-wall,

insulated, pressurized plenum equipment enclosures shall be provided as indicated on the Drawings.

B. Panel Construction:

1. Panels shall be 4 inches thick with minimum 20 gauge exterior solid panel and 22 gauge perforated interior panel with 3/32 inch holes with maximum 23 percent open area. Minimum open area shall be 22 percent.
2. All channel members shall be structural quality galvanized steel and a minimum of 18 gauge.
3. Panel materials shall be as specified for ductwork.
4. Panels shall be welded to structural members as required to meet pressure and structural performance requirements.
5. Panel assembly shall be filled with inert, mildew-resistant and vermin-proof acoustical/insulating materials. Flame spread rating shall not be greater than 25 and smoke developed rating not greater than 50 per ASTM E-84 and as specified by ANSI/NFPA 90A.
6. Panels shall be tongue and groove construction with adjacent panels held together with continuous self locking joint.

C. Access Doors:

1. Doors shall be 24 inches wide by 60 inches high unless noted otherwise.
2. Door panels shall be solid panel on each side and same thickness as wall panels.
3. Each door shall have a minimum of 2 ball bearing hinges and 2 wedge-lever door latches operable from both sides.
4. Doors shall seat against full perimeter neoprene gasket, and arranged to open against pressure differential.

D. Structural Performance:

1. The entire plenum shall be designed to be self-supporting. When additional structural strength is required heavier panel skins and/or structural reinforcing shall be provided.
2. Under operating conditions maximum panel joint deflection shall not exceed $L/200$ where L is the maximum unsupported span length of any panel section.
3. Minimum static pressure rating shall be as specified Section 15840, 2.3.

E. Minimum Acoustical Performance Shall be as Follows:

Octave Band	2	3	4	5	6	7
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Absorption Coefficient	0.89	0.99	0.99	0.99	0.99	0.96
Transmission Loss	22	28	38	48	54	58

1. Test results shall be certified by an independent testing agency and made in accordance with ASTM Standards E 90 and E 413.

2.6 ACOUSTICAL DUCT

- A. Provide where shown on the Drawings.
- B. Construction:
 1. Round duct with spiral lockseam construction.
 2. Outer shell to be solid with perforated inner liner.
 3. Provide acoustical material between inner and outer shells with a fiberglass cloth between inner shell and acoustical material. Acoustical material shall be 1 inch or 3 inches in depth as shown on Drawings.
 4. Pressure rating shall be 2 inches.
- C. Product shall be United McGill Acoustic-K27 round duct or equal.
- D. Outer shell shall comply with 15840, 2.2C when buried.

2.7 SEISMIC DUCT JOINT

- A. Where shown on the Drawings and/or required due to building structural separations, provide flanged rubber joints made of chlorobutyl elastomer with polyester reinforcing fabric rated for 300 degrees F operating temperature, 3 psi air pressure, and 83 inches H2O vacuum.
- B. Use General Rubber Corporation Series 3000, Style 1097 joints on rectangular duct and Style 1098 double bellow on round duct.

PART 3 - EXECUTION

3.1 FABRICATION

- A. General: Fabricate ductwork in accordance with applicable SMACNA HVAC and Industrial Duct Construction Standards and ASHRAE handbooks unless more stringently specified or detailed.
 1. Construct ductwork to comply with Section 15840, Articles 1.3 and 1.4.

2. Ductwork which is fabricated and installed without coordination is subject to removal and replacement at contractors cost. Coordination drawings are required to be prepared in accordance with Section 01310.
 3. Fabricated ductwork sections and fittings shall have an identification sticker applied at fabrication shop. Sticker shall indicate size and metal gauge of component.
 4. Fabricate kitchen hood exhaust ducts with carbon steel sheets for concealed ducts and stainless steel for exposed ducts. Weld and flange seams and joints.
 5. Fabricate dishwasher hood exhaust ducts with 18 gage stainless steels. Weld and flange seams and joints.
- B. Minor variation of duct configuration or size is permitted provided system resistance does not increase. See Article 3.1.L.
- C. Complete metal ducts within themselves with no single partition between ducts. Open corners are not acceptable.
- D. Lap metal ducts in direction of air flow. Hammer down edges and slips to leave smooth duct interior.
- E. Fabricate tees, bends, and elbows with radius of not less than 1-1/2 times width of duct on center line. Where rectangular elbows are used, provide turning vanes. All mitered elbows shall have single thickness turning vanes; dual wall airfoil type blades are not acceptable. Turning vanes shall be secure and stable in track. Track section shall be attached to fitting on each side with 2 sets of screws or bolts.
- F. Increase duct sizes gradually, not exceeding 15 degrees divergence wherever possible. Maximum divergence upstream of equipment to be 30 degrees and 45 degrees convergence downstream.
- G. Rigidly construct metal ducts with joints mechanically tight, substantially airtight, braced and stiffened so as not to breathe, rattle, vibrate, or sag. Seal duct joints and connections with sealant as ducts are being assembled.
- H. Provide necessary baffling in mixed air plenums to ensure good mixed air temperature.
- I. Fabricate round and oval duct fittings with spiral or continuously welded seams. Joints shall be 2 inches slip type, cemented, brazed, electric welded, or with shrinkbands. Prime coat welded joints. Fabricate elbows

of 5-piece construction. Provide standard 45 degree takeoffs unless otherwise indicated.

- J. Reinforce plenum or casing door frames with galvanized steel angles tied to horizontal and vertical plenum supporting angles. Install hinged access doors for access to equipment, for cleaning, and inspection.
- K. Minimum size of access doors into plenums shall be 24 inches W by 60 inches H unless dimensioned otherwise on Drawings. Doors shall swing to seat with pressure gradient.
- L. Coordinate ductwork systems layout with other trades prior to fabrication. Provide offsets, transitions and stream-liners as required to clear structural interferences and mechanical or electrical systems. Working plans do not indicate all fittings, offsets, transitions and stream-liners that are required to install ductwork systems.
- M. Balancing dampers are required at every branch of all ductwork systems. Use of O.B.D.s in ceiling diffuser outlet, shall not be used for systems balance.

3.2 INSTALLATION

- A. Duct System Pressure Class: Construct and install each duct system for the specific duct pressure classification indicated.
- B. Provide openings in ductwork to accommodate thermometers and controllers. Provide pivot tube openings for testing of systems, complete with metal cam with spring device or screw to ensure against air leakage. Where openings are provided in insulated ductwork, install insulation material inside a metal ring.
- C. Install ducts with the fewest possible joints.
- D. Use fabricated fittings for all changes in directions, changes in size and shape, and connections.
- E. Install couplings tight to duct wall surface with projections into duct at connections kept to a minimum.
- F. Locate ducts, except as otherwise indicated, vertically and horizontally, parallel and perpendicular to building lines; avoid diagonal runs. Install duct systems in shortest route that does not obstruct useable space or block access for servicing building and its equipment.

- G. Mount duct-type smoke detectors on the ductwork under this Section of the Specifications. All mounting arrangements, holes cut into ductwork, sealing of openings along with ceiling and access doors for each duct-type detector shall be provided under this Section of the Specifications. Duct detectors along with all sampling tubes and accessories shall be furnished under Division 16.
- H. Clean ductwork as it is installed to remove dirt and dust. During installation, close duct openings and open ends with temporary covers to keep construction dust out of duct systems.
- I. Construct ductwork with sufficient clearance around equipment to allow normal operating and maintenance activities of equipment.
- J. At each point where ducts pass through partitions, provide sleeves and seal joints around duct with noncombustible material. Refer to Section 15140.
- K. Where manufactured duct joint systems such as "Ductmate" are employed, install components in strict accordance with the manufacturer's recommendations. Bolt metal corner pieces. PVC clips are not allowed.
- L. Install ducts and supports in accordance with SMACNA Standards. Adequately attach hangers to the building structure. Refer to Sections 15245 and 15140. For seismic joint crossings, refer to Section 15140. Vibration isolate ductwork as specified in Section 15240. Seismic restraint shall not short-circuit vibration isolation systems.
- M. Prior to operation or test and balance, clean duct systems with high power vacuum machines. Protect equipment which may be harmed by excessive dirt, with filters or bypass during cleaning. Provide adequate access into ductwork for cleaning purposes spaced no more than 50 feet on center.
- N. Shrink-Fit Coupling: Where specified, use joints in strict accordance with manufacturer's instructions. Raychem Thermofit wrap or equal.
- O. Temporary sheet metal screws are not permissible at fire or smoke damper breakaway joints.
- P. Ducts shall not be insulated prior to observation of sealed joints and for duct leak testing.

- Q. Set plenum doors 6 inches to 12 inches above floor. Arrange door swings so that fan static pressure holds door in closed position.

3.3 INSTALLATION ACOUSTICAL PLENUMS

- A. Provide a 4 inch curb upon which to set panels. Coordinate with Division 3.
- B. Install plenum base channel on level concrete curb.
- C. Installation of enclosure shall be in accordance with manufacturer's instructions.
- D. All mechanical joints and external trim items shall be sealed.
- E. Locate access doors for each compartment and as shown on Drawings.
- F. Openings for pipe and conduit penetrations shall be field cut. Provide flashing and seals per manufacturer's detail.

3.4 INSTALLATION OF BURIED DUCTWORK

- A. Placement of all buried ductwork shall be fully coordinated with Structural and Architectural Drawings to insure location of duct penetration through floor slabs is free of offsets.
- B. All buried ductwork shall be insulated as specified in Section 15290. Insulation shall be applied to duct sections and covered prior to section placement in soil. Insulation of joints between sections may be applied after sections are made-up in trench. All insulation shall be covered prior to backfilling activity.
- C. Duct sections shall be backfilled uniformly. Duct dimensional stability will be maintained throughout. Backfilling materials and procedures shall be in accordance with duct manufacturer's recommendations and in compliance with SMACNA guidelines.
- D. Annular space between outer duct insulation and opening through floor slabs shall be 1 inch plus or minus 1/2 inch.

3.5 QUALITY ASSURANCE

- A. Maximum Allowable Leakage: As described in ASHRAE 1993 Handbook, "Fundamentals" Volume, Chapter 32, Table 7 and

Figure 10. Comply with requirements for leakage classification 3 for round and flat oval ducts, leakage classification 12 for rectangular ducts in pressure classifications less than and equal to 2 inches water gage (both positive and negative pressures), and leakage classification 6 for pressure classifications greater than 2 inches water gage and less than and equal to 10 inches water gage.

- B. Remake leaking joints as required and apply sealant to achieve specified maximum allowable leakage.
- C. Leakage Test: Perform volumetric measurements and adjust air systems as described in ASHRAE 1995 "HVAC Applications" Volume, Chapter 34, and Division 15 Section TESTING, ADJUSTING, AND BALANCING.
- D. Owner's Representative shall witness all leakage tests with 48 hours notice provided before any leak test or retest is conducted.
- E. All leakage tests shall be documented and submitted for record within 72 hours after testing. Ductwork shall not be fully concealed or made inaccessible for repair until after Owner acceptance of leak test report.

- END -